(8) Correlations

PAGE NO.....
DATE / /

→ Correlation measures the relationship between two variables. It shows how changes in one variable are associated with change in another.

(1) Types of Correlation (Based on

(A) Based on Direction.

↳ (1) Positive Correlation
  ↳ Both variables increases together
  ↳ E.g: study time & Exam scores

↳ (2) Negative Correlation.
  ↳ One variables increases while the other decreases
  ↳ E.g: price of goods & Demand.

↳ (3) No correlation
  ↳ No relationship between variables.
  ↳ Shoe size & IQ.

(B) Based on Strength.

Pearson's correlations coefficient (r) interpretation:

r	strength.
+1.0 to +0.7	Strong positive.
+0.7 to 0.3	Moderate positive.
+0.3 to 0.0	Weak positive.
0.0	No correlation.
-0.3 to -0.7	Moderate negative.
-0.7 to -1.0	Strong negative.

(C) Based on Computation Method.

↳ (1) Pearson's Correlation (Parametric)
↳ (2) Spearman's Rank Correlation (Non-Parametric)
↳ (3) Kendall's Tau Correlation (Non-Parametric)

PAGE NO.....
DATE / /

1) Pearson's Correlation?

→ Pearson's correlation coefficient (r) measures the linear relationship between two continuous variables.

→ It tell us
  ↳ How strong the relationship is
  ↳ Whether the relationship is positive or negative.

→ The coefficient r ranges from -1 to +1
- +1 → Perfect positive correlation (↑↑ or ↓↓)
- -1 → Perfect negative correlation (↑↑)
- 0 → No correlation (random scatter).

(2) Mathematical Approach.

$$r = \frac{\sum (x-\bar{x})(y-\bar{y})}{\sqrt{\sum (x-\bar{x})^2} \cdot \sqrt{\sum (y-\bar{y})^2}}$$

where.

→ x, y → The two variables (e.g. Height & weight).

→ $\bar{x}, \bar{y}$ → Mean (average) of x and y

→ $\sum$ → summation

→ $(x-\bar{x})$ → Deviation of each value from the mean.

→ $(y-\bar{y})$ → Deviation of y values from their mean.

Given

Person	A	B	C	D	E
Height (cm)	160	170	175	180	185.
weight (kg)	58	65	72	80	85

DATE / /

(a) Mean of height ($\bar{x}$) = 174

(b) Mean of weight ($\bar{y}$) = 72

→ We can calculate manually, let's jump into python:

(1) Manual calculation in Python.

```
import numpy as np.

# Data
height = np.array([160, 170, 175, 180, 185])
weight = np.array([56, 65, 72, 80, 85])

# calculate mean
mean_height = np.mean(height)
mean_weight = np.mean(weight)

# Calculate Pearson's correlation manually
numerator = np.sum((height - mean_height) *
                   (weight - mean_weight))

denominator = np.sqrt(np.sum((height - mean_
                height)**2) * np.sum((weight -
                mean_weight)**2))

r = numerator / denominator

print(f"Pearson's correlation coefficient: {r:.2f}")
```

output

0.98

PAGE NO.....
DATE / /

(2) Using pandas. DataFrame.corr() (Simplest).

```
import pandas as pd.

# Sample data
data = {'Height': [160, 170, 175, 180, 165]
        'Weight': [55, 65, 72, 80, 60]}

df = pd.DataFrame(data)

# Person correlation
pearson_corr = df.corr(method='pearson').
    print(pearson_corr)
```

-> Automatically handles missing values.
-> Works for large datasets.

(3) Using numpy.corrcoef() (For simple arrays).

```
import numpy as np.

#Convert lists to NumPy arrays.
height = np.array([160, 170, 175, 180, 165])
weight = np.array([55, 65, 72, 80, 60]).

# Compute Pearson correlation
Corr_matrix = np.corrcoef(height, weight).

print(corr_matrix)
```

-> Fastest for simple numerical arrays
-> Doesn't compute p-values.

PAGE NO.....
DATE / /

(4) Using scipy.stats.pearson() (Gives p-value too)

⇒ This function returns both the correlations coeff. and the p-value, which helps in checking statistical significance.

```
from scipy.stats import pearsonr

# Sample data
height = [160, 170, 175, 180, 165]
weight = [55, 65, 72, 80, 60]

# Compute Pearson correlation and p-values.
r, p-value = pearsonr(height, weight).

print(f"{r:.3f}")
print(f"{p-value:.3f}").
```

where

- p-value tells us if a correlation is statistically significant
- if p-value < 0.05, we reject the null hypothesis (strong evidence of correlation).
- if p-value > 0.05, the correlation might be random (not significant).

PAGE NO.....
DATE / /

(2) Spearson's Rank Correlation (Non-Parametric) - 📝

→ What is Spearman's Rank Correlation?

↳ Spearman's correlation is used to measure the strength and direction of a monotonic relationship between two variables. Unlike Pearson's correlation (which measures linear relationships), Spearman works with ranked data and is useful when the data isn't normally distributed or has non-linear trends.

Formula:

$$r_s = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

where,

d = difference between the ranks of two variables.

n = Number of observations.

→ The formula essentially checks how far each ranking is from the expected perfect rank order. A smaller d^2 means a stronger correlation.

When to use?

→ Your data is ordinal (like ratings, rankings, survey scores)

→ The relationship is non-linear but still monotonic (as one variable increases, the other consistently increases or decrease)

→ Your data is not normally distributed, meaning Pearson's correlation might not work well.

DATE / /

Let's say we want to find the relationship between:
1) product Price (in dollars).
2) Customer Satisfaction (1 to 5 stars).

Even if there's no perfect linear relationship, Spearman's correlations can still tell us whether higher prices tend to get better (or worse) reviews.

(1) Manual calculation.

product	Price ($)	Satisfaction (stars)	Rank (price)	Rank (satisfaction)	Rank diff	d^2
A	20	5	5	1	4	16
B	40	3	4	3	1	1
C	10	4	1	2	-1	1
D	30	2	3	4	-1	1
E	50	1	2	5	-3	9

Now, using the formula

$$r_s = 1 - \frac{6 \sum d^2}{n(n^2-1)}$$

$$= 1 - \frac{6(16+1+1+1+9)}{5(5^2-1)}$$

$$= 1 - \frac{6(28)}{5(24}$$

$$= 1 - \frac{168}{120}$$

$$= 1 - 1.4 = -0.4$$

This means as price increases, customer satisfaction tends to decrease (negative correlation).

PAGE NO.....
DATE. / /

(2) Python

```
import scipy.stats as stats.

prices = [20, 40, 10, 20, 50]  # prices of product.
Satisfactions = [5, 3, 4, 2, 2]  # Customer rating
                                   (1-5 stars).

# Compute Spearson's correlations.

spearman_corr, p_value = stats.spearmanr(prices,
                                         Satisfactions).

print(f"Spearsons correlations coefficient:
      {spearman_corr:.3f}").

print(f"p-value : {p_value:.3f}")
```

Interpreting the results

- If $r_s \approx 1$ -> Strong positive correlation (higher price = higher satisfactions).
- If $r_s \approx -1$ -> Strong negative correlation (higher price = lower satisfactions)
- If $r_s \approx 0$ -> No correlation (price and satisfaction are unrelated)

PAGE NO.....
DATE / /

Pair wise Comparisons

Employee Pair	Experience Order	Performance Order	concordant or Discordant
(A, B)	3 < 5	2 < 4	concordant
(A, C)	3 > 1	2 > 1	"
(A, D)	3 < 4	2 < 3	"
(A, E)	3 > 2	2 < 5	Discordant
(B, C)	5 > 1	4 > 1	concordant
(B, D)	5 > 4	4 > 3	concordant
(B, E)	5 > 2	4 < 5	Discordant
(C, D)	1 < 4	1 < 3	concordant
(C, E)	1 < 2	1 < 5	"
(D, E)	4 > 2	3 < 5	Discordant.

(2)

$$\gamma = \frac{C - D}{\frac{1}{2} n(n-1)}$$

$$= \frac{7 - 3}{\frac{1}{2}(5 \times 4)}$$

$$= \frac{4}{10} \geq 0.4$$

→ A $\gamma \geq 0.4$ means there is a moderate positive correlation between experience and job performance.

PAGE NO.....
DATE / /

python

```
import scipy.stats as stats

# Sample Data
experience = [2, 5, 1, 4, 2] # Year of experience
performance = [2, 4, 1, 3, 5] # Job performance ranking

# Compute Kendall's Tau correlation
tau, p-value = stats.kendalltau(experience,
                                performance)

print(f"Kendall's Tau coefficient: {tau:.3f}")
print(f"p-value: {p-value:.3f}").
```

Interpretation of Kendall's Tau:

- if $\tau \approx 1$ -> Strong positive correlation (as experience increases, performance increases)
- if $\tau \approx -1$ -) Strong negative correlation (as experience increases, performance decreases)
- if $\tau \approx 0$ -) No correlation (experience and performance are unrelated)

DATE / /

Pearson vs. Spearman vs. Kendall's Tau

(1) What do they Measure?

Correlation type	Measure	Best for
Pearson's (r)	Linear relationship between two continuous variables	when variable have a linear rel^n. and data is normally distributed
Spearman's (ρ)	Monotonic relationship based on ranked values.	when data is ordinal or does not follow a normal distribution
Kendall's Tau (τ)	Monotonic relationship based on paired ranking.	small datasets with tied ranks.

(2) Key difference

Feature	Pearson's (r)	Spearman's (ρ)	Kendall (τ)
Data type	Continuous (numeric)	Ordinal or Ranked	Ordinal or Ranked.
Assumpt^n	Data must be normally distributed.	No normality assumption	No normality assumption.
Handles Outliers?	very sensitive to outlier	less sensitive	Less sensitive.
Works with Tied Ranks?	No	Partially	Best.
Scalability (large Data)	works well	works well	Slower for Large datasets.
Best for Dataset	Large	Large	Large